Northeastern University
Bouvé College of Health Sciences

Course number: PHTH 6802

Course name: Causal Inference 2

Instructor

Instructor:
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Office hours: [day and time TBD] via Zoom. Email me to arrange another time, or to meet in person (Wednesdays only).

Course Information

Description:

Continues PHTH 6801. Expands on foundational knowledge of causal inference by examining time-varying exposures, introducing the g-formula for estimating standardized outcome distributions, and unraveling the intricacies of marginal structural models. Navigates through key topics, including static and dynamic treatment regimes. Engages in discussions on sensitivity analysis, graphical models, identification algorithms, and the complex domain of causal discovery. Examines advanced techniques in causal inference, offering students an opportunity to apply theoretical principles to practical scenarios. Tackles challenging aspects such as time-varying exposures and sophisticated modeling techniques in the pursuit of accurate and meaningful outcomes.

Prerequisites:

PHTH 6800 with a minimum grade of B or PHTH 6801 with a minimum grade of B

Corequisites:

None.

Credit Hours:

4

Objectives / Outcomes

At the end of this course, you will be able to meet the following outcomes:

- Use tools such as causal diagrams and counterfactual theory to pose causal questions about treatments that vary over time

- Choose among and apply g-methods to answer causal questions that standard analytic methods cannot address
- Assess the validity of a longitudinal study's design and analysis, and whether the effect it estimates is the effect of interest
- Design and analyze a target trial emulation to answer a causal question

Teaching Methods / Learning Experiences

The course meets once a week. Each section will be a mix of: lecture, reading and discussion, individual and group problem solving, and project work.

Grading / Evaluation

Assignment	% of grade
Participation	30%
In-class mini-quizzes and reading reflections	30%
Midterm (tentative date: October 21)	20%
Final exam (December 16)	20%

Mini-quizzes will generally consist of a sample of questions similar or identical to ungraded problem set questions. Reading reflections will be in-class written responses to articles assigned the previous week.

Grading scale:

A 100% to 93.0%
A- 92.9% to 90.0%
B+ 89.9% to 87.0%
B 86.9% to 83.0%
B- 82.9% to 80.0%
C+ 79.9% to 77.0%
C 76.9% to 73.0%
C- 72.9% to 70.0%
F Below 70.0%

Rounding policy: Final grades will be rounded *up* to the nearest tenth of a percent.

Materials / Resources

Required Text:

Hernán MA, Robins JM (2020). Causal Inference: What If. Boca Raton: Chapman & Hall/CRC

This book is freely available online: https://miguelhernan.org/whatifbook

Articles from the methodological and applied literature will be assigned both in and out of class as required and supplemental readings.

Canvas: https://canvas.northeastern.edu

All course materials will be posted to Canvas except the textbook. Announcements are sent through Canvas, so please make sure your notification settings deliver them somewhere you will see them.

Course Policies

Student Expectations:

Bring a pen/pencil to class for in-class assignments. You will also likely prefer a pen and paper (or tablet and stylus) for taking notes, which will include equations and graphical models. I discourage taking notes on a laptop, both because learning is better done through handwriting, and to avoid distractions. If you are using a device with a screen visible to others, it must only show your notes (no messaging, shopping, etc.) to avoid distracting your classmates.

Skim or read the assigned textbook sections before the session they support. You don't need to understand it fully until after the weekly session and you get a chance to practice the content, but come prepared to ask questions. Required article readings will be accompanied by a specified due date that they should be completed before.

Attendance is expected. The majority of graded work will be conducted in class.

Class Participation:

This is a participatory course and each student is expected to be an active participant in the work that occurs. Student learning in this course is directly related to the efforts made in learning the course material. Attending class and arriving to class promptly is expected of each student (and shows respect for fellow students). Students will be called on in class, and each student can expect to be called on unexpectedly several times during the semester.

Professional Conduct:

Part of the student learning experience is to practice professional behavior. This includes affording your student peers, presenters and professors the courtesy of not using computers, cell phone, pagers or text messaging during class. Non-adherence to these expectations of professional behavior will result in dismissal from class. Consistent with University policy, students must seek DAS permission to record class lectures – please see university policy on recording.

Make Up Policy:

The midterm and final are given in class on the dates in the schedule. If you have a documented emergency, a religious observance, or a university-sanctioned obligation that conflicts with an exam, contact me in advance and we will arrange an alternative.

Your lowest in-class assignment score will be dropped automatically, which is intended to cover one absence without any need to explain it. If you miss more than one session, get in touch to arrange a make-up plan.

Late Work Policy:

Submitted work is generally done during class so will not be accepted late, though I reserve the right to move in-class assignments out of class as necessary. In such cases, work submitted late for any reason other than one discussed with me ahead of time will lose 20% per day.

If something in your life is making a deadline impossible, tell me before the deadline rather than after it. I am considerably more flexible in advance than in retrospect.

Academic Integrity:

As health professionals and scientists-in-training, the highest level of ethical behavior is expected of all students. Plagiarism, cheating, and any form of unauthorized collaboration will not be tolerated and will be handled in accordance with University policies including the Academic Honesty and Integrity Policy. Requirements can be found at: https://osccr.sites.northeastern.edu/academic-integrity-policy/

https://osccr.sites.northeastern.edu/code-of-student-conduct/

- Acts of academic dishonesty will be referred to OSCCR (Office of Student Conduct and Conflict Resolution).
- Infractions will be dealt with according to Northeastern Disciplinary Process shown in http://www.northeastern.edu/osccr/academic-integrity-report-form/
- The Turnitin application may be used by the course instructor to identify and document cases of plagiarism.

Use of Generative Artificial Intelligence (AI)

Please review Northeastern's Policy on the Use of Artificial Intelligence Systems.

Outside of class, you may use AI tools if you wish to explain or check your understanding of material, conduct lit reviews, or other tasks that help you learn. However, this comes with the strong caveat that the material this course covers is not always well-documented in the literature and is not necessarily intuitive, and AI **often** gets it wrong in ways that it doesn't get wrong, say, basic biostatical concepts. You are ultimately responsible for what you learn, and the materials provided in class will be much more trustworthy (with the necessary caveats about the possibility for errors in the literature, and certainly for typos and other errors in my materials) than anything AI can provide.

The in-class assignments are designed to test what you have understood on your own, without AI. Any out-of-class assignment will come with a specific AI policy.

Version 021825

I reserve the right to change this overall policy, as well as any aspect of the syllabus (with appropriate warning), due to concerns about AI use or otherwise.

Belonging at Bouvé College:

We acknowledge that chronic and persistent disparities in education and health due to systemic oppression, based on race, class, and other marginalized identities, have adversely affected individuals and society. At Bouvé we are committed to educating future healthcare leaders who strive to serve marginalized communities and improve the well-being of all members of society through the health span.

Within the college, we strive to create a safe environment for diverse thoughts, perspectives, and experiences to be shared, explored, and embraced. As a community we recognize that disrupting long-standing oppressive practices is a process. Everyone is at a different place on this journey, and we affirm our commitment to support and learn from one another.

In order for everyone in our learning community to grow, gain a deeper understanding of complex issues and be exposed to diversity of thought, we may have discussions in this class on topics that could be uncomfortable or difficult for some people. It is imperative that we follow rules of engagement as below in order to create a psychologically safe learning environment.

Ground Rules:

1. Listen actively and respect others when they are talking.
2. Participate to the fullest of your ability without taking up too much space – community growth depends on the inclusion of every individual voice
3. Speak from your own experience instead of generalizing ("I" instead of "we")
4. Do not be afraid to respectfully challenge one another by asking questions but refrain from personal attacks and stay focused on ideas rather than individuals
5. Be conscious of body language and nonverbal responses; they can be as disrespectful and hurtful as words.

Finally, all community members need to adhere to the NU Student Code of Conduct "e*ach Northeastern student has the responsibility to respect the differences of individuals and treat others in a civil and respectful manner."*

This course draws its examples from real public health research, and some of them are not neutral: racial and ethnic disparities in access to services, disability, screening programs and who gets missed by them, and clinical trials conducted in resource-limited settings. We will treat those examples as what they are, which is data about people's lives. All class and group work interactions are expected to be courteous and respectful.

I consider it part of my job to address the learning needs of every student in this course, and the range of preparation people bring to a first statistics class is wide. If something about how this course runs is making it harder for you to learn, tell me and I will try to fix it.

Belonging at Northeastern

Students with Disabilities:

Disability Access Services (DAS), located on campus in 10 Dodge Hall (617-373-2675) can provide students with information and other assistance to help manage any challenges that may affect their performance in the coursework. The University requires that students provide documentation of their disability to the DAS. Students should meet with the course instructor for special accommodations to be arranged.

Northeastern University abides by Section 504 of the Rehabilitation Act of 1973, which stipulates that no student shall be denied the benefits of an education "solely by reason of a handicap." Disabilities covered by law include but are not limited to, learning disabilities and hearing, sight or mobility impairments. Additional information about DAS can be found online at https://disabilityaccessservices.northeastern.edu/

Student Wellness:

As health researchers, we do our best to maintain a healthy, balanced life. As a student however, you may experience a range of challenges including significant stress, difficult life events, mood changes, excessive worry, or problems with eating and/or sleeping. These can diminish your academic performance and/or reduce your ability to participate in daily activities. If you or anyone you know is struggling, we strongly encourage you to seek support. Northeastern University provides several services and resources to support the overall wellness of students:

University Health and Counseling Services

Find at Northeastern

We Care

Wellness Days

As always, please contact Northeastern University Police Department at 617-373-3333 in emergency situations.

Title IX Policy:

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including students, faculty and staff of all gender identities.

Learn more at:

https://www.northeastern.edu/ouec/training-and-education/syllabus-language-title-ix/

Know Your Options: Pregnancy

Course Feedback:

The Office of the Registrar, in collaboration with the Office of the Provost, administers the TRACE (Teacher Rating and Course Evaluation) for undergraduate and graduate courses at Northeastern University. Students' assessments through TRACE are completely anonymous. Given the public nature of student comments made through TRACE and their intended use to enhance teaching, students are asked to please be thoughtful, professional and considerate with responses.

Syllabus Disclaimer:

Any changes to the syllabus will be made at the discretion of the instructor. It is the responsibility of the student to stay informed of any such changes.

For additional information, resources, and policies please visit the Northeastern University Catalog* including:

- University Undergraduate and University-wide academic policies and procedures
- University Graduate and University-wide academic policies and procedures
- Bouvé College of Health Sciences Undergraduate and Graduate policies, and program-specific policies and procedures

**Students should consult the appropriate version based on enrollment of the Northeastern academic catalogs, their academic advisors, and/or their degree audit reports for their specific requirements.*

Major Assignment / Course Schedule

Topics and dates subject to change based on timing and student/instructor interest.

Weeks/Dates	Topics/Assignments/Assessments/etc.	Due Dates
Week 1 Wed Sep 9	PHTH 6800 review, fixed vs. time-varying exposures and confounders, notation, treatment strategies, generalized identification assumptions	
Week 2 Wed Sep 16	Sequentially randomized experiments; SWIGs for time-varying treatments; treatment–confounder feedback and why standard methods fail	In-class assignment
Week 3 Wed Sep 23	Tree graphs; the g-formula for static strategies; the ICE representation	In-class assignment
Week 4 Wed Sep 30	IP weighting, pseudopopulations, MSMs (saturated vs. unsaturated); equivalence with the g-formula; double robustness	In-class assignment
Week 5 Wed Oct 7	Dynamic strategies; SWIGs for dynamic strategies; dynamic MSMs; optimal regimes; testing causal assumptions from the data	In-class assignment
Week 6 Wed Oct 14	Censoring as a time-varying treatment; survival outcomes and pooled logistic regression; comparing	In-class assignment

Weeks/Dates	Topics/Assignments/Assessments/etc.	Due Dates
	the g-methods; g-null paradox; g-estimation and SNMs	
Week 7 Wed Oct 21	Midterm (first half of class) Target trial emulation I: the framework; protocol components; ITT vs. per-protocol effects	Midterm [tentative date]
Week 8 Wed Oct 28	Target trial emulation II: eligibility, time zero, and immortal time; single vs. multiple time zeros; prevalent-user and other alignment failures	In-class assignment
Week 9 Wed Nov 4	Clone-censor-weight; grace periods as dynamic strategies	In-class assignment
Week 10 Wed Nov 11	**NO CLASS — Veterans Day** Reading completed on your own	
Week 11 Wed Nov 18	Sequential trial emulation; adherence; estimating per-protocol effects	In-class assignment
Week 12 Wed Nov 25	**NO CLASS — Thanksgiving break**	
Week 13 Wed Dec 2	Competing events	In-class assignment
Week 14 Wed Dec 9	Separable effects and interventionist mediation	In-class assignment
Week 15 Wed Dec 16	**FINAL EXAM (full session, cumulative, but more emphasis on content since midterm)**	Final exam, in class